

17

Light is polarised when it passes through a sheet of material known as polaroid.

Two sources producing coherent light waves are placed at an equal distance away from an observation screen. Each source was covered with a polaroid.

The diagram illustrates a double-slit interference experiment. On the left, two small circles represent 'coherent light sources'. Each source is positioned behind a vertical rectangular block labeled 'polaroids'. A horizontal dotted line extends from the midpoint between the two sources to a vertical line on the right labeled 'screen', representing the central axis of the experiment.

Initially, both polaroids had their transmission axes in the same direction. The intensity of the central maximum fringe formed from the interference of the two light waves was measured to be I .

One of the polaroids is rotated by 60° .

What is the new intensity of the central maximum fringe?

A

$$\frac{1}{2}I$$

B

$$\frac{9}{16}I$$

C

$$\frac{3}{2}I$$

D

$$\frac{9}{4}I$$

Answer: B

Initially, the intensity of the central maximum fringe is I .

$$\begin{aligned}
 I &= k(A_1 + A_2)^2 \\
 &= k(2A)^2 \\
 &= 4kA^2
 \end{aligned}$$

After the rotation of the polaroid,

$$\begin{aligned}
 I_{new} &= k(A_1 + A_2)^2 \\
 &= k(A + A \cos 60)^2 \\
 &= k\left(\frac{3A}{2}\right)^2 \\
 &= \frac{9}{4}kA^2 \\
 &= \frac{9}{16}(4kA^2) \\
 &= \frac{9}{16}I
 \end{aligned}$$

18 The diagram shows a stationary wave of frequency 50 Hz formed between two points P and Q.